Neel Gali

Software Quality Engineer

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | AI-Powered Test Automation Platform

- Architected and implemented an AI-driven test script generation framework utilizing **Python**, **LangGraph**, and **OpenAI API**, improving test case creation efficiency by 40%.
- Led the integration of **Large Language Models (LLMs)** into existing automation suites, using **CrewAI** agents for intelligent test data generation and scenario exploration, reducing manual test data setup by 30%.
- Designed and deployed scalable test infrastructure on **Kubernetes** clusters, orchestrating **Docker** containers for parallel test execution, boosting regression testing throughput by 50%.
- Developed advanced **Python** utilities for dynamic log analysis and anomaly detection, leveraging **AI tools** to pinpoint **root causes** in complex distributed systems, accelerating debugging by 25%.
- Mentored junior engineers on best practices for developing high-quality, maintainable **test scripts** and automation frameworks, improving team code quality metrics by 20%.
- Enhanced system **quality and regression** capabilities by implementing predictive defect identification models using **Python** and machine learning algorithms, reducing critical bug escapes by 15%.
- Collaborated with product teams to translate new feature requirements into robust **test plans**, ensuring comprehensive validation against design specifications and improving feature readiness by 35%.
- Orchestrated continuous integration of automated tests within **GitLab CI**, establishing robust gating mechanisms for code merges, thereby enhancing overall code stability by 20%.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | Enterprise Network Solutions QA

- Led automation of complex integration tests for multi-service network applications using **Python** and **Robot Framework**, achieving 95% test coverage for critical features.
- Deployed and managed containerized test environments using **Docker** and **Kubernetes** to simulate large-scale network topologies, improving test environment setup efficiency by 50%.
- Developed custom **Python** scripts for validating network device configurations, including **STP** and **ISIS** protocols, ensuring compliance and reducing configuration errors by 30%.
- Implemented **API testing** strategies for RESTful services using **Postman** and **Python**'s 'requests' library, identifying integration defects early and accelerating development cycles by 20%.
- Streamlined defect reporting and tracking processes using **Jira** and integrated automated alerts, enhancing developer-QA collaboration and reducing bug resolution time by 15%.
- Explored **Large Language Model (LLM)** capabilities by developing a **Python** prototype using **OpenAI API** for generating initial test case drafts, reducing manual documentation effort by 10%.
- Maintained **Linux**-based test servers and performed system-level debugging, leveraging advanced shell scripting to diagnose performance bottlenecks, improving test execution stability by 25%.
- Contributed to **CI/CD** pipeline enhancements using **Jenkins** and **Git** for automated build verification, enabling faster feedback loops and improving code deployment frequency by 2x.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Cloud-Based Platform QA

- Designed and executed comprehensive **test plans** for critical cloud-based applications, employing **Python** scripting and **Robot Framework** to validate features across releases, improving test coverage by 35%.
- Developed robust automated **test scripts** using **Python** and **Selenium** for web-based UI testing, integrating with **Jenkins CI/CD** pipelines to reduce manual testing efforts by 40%.
- Performed in-depth **root cause analysis** on identified software defects, filing detailed **Jira** tickets and collaborating with developers to achieve a bug resolution rate of 90%.
- Configured and managed virtualized test environments on **Linux** servers, applying knowledge of **VLAN** and **DHCP** to simulate network conditions, enhancing test environment realism by 25%.
- Validated network protocols like **OSPF** and **BGP** within embedded systems using specialized tools, ensuring device interoperability and stability for enterprise clients, reducing critical network issues by 15%.
- Maintained and enhanced existing **regression test suites** for legacy systems, streamlining test execution with **Bash** scripts on **Linux**, accelerating release cycles by 20%.

Skills

Languages: Python, Bash
Frameworks & Tools: Selenium, Jira, Jenkins, Postman, Git, GitLab C, AI Tools, OpenAI API, LangGraph, CrewAI
Cloud & Infra: Linux, Kubernetes, Docker, VLAN, DHCP, OSPF, BGP, STP, ISIS
Concepts: CI/CD, Test Automation, Regression Testing, Root Cause Analysis, API Testing, Test Plans, LLMs

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India